

Unit - 1 Introduction to Python.

2 Mark Questions.

1) Who created python and in which year was it first released?

- Python was created by Guido Van Rossum and was first released in the year 1991.
- The first official version, python 0.9.0, was released on february 20, 1991.

2) Mention any two features of python.

The Two Key features of python are:

1. Simple and Readable System
2. Interpreted and High-level language

3) What are Keywords in python? Give two Examples.

Keywords in python are reserved word that have a special meaning and purpose in the language

- 3) Cannot use python Keywords:
Examples of Keywords : class, if, while, def, for
- Valid : My-class
 - Invalid class (it's a keyword)

- 4) Case - Sensitive:
name, Name and NAME are treated as different identifiers.

- 5) Should not contain special characters.
Such as : @, #, \$, % etc.
Invalid : User@name, total%value.

4) Write Short notes on python data types with suitable examples.

Python has Various built-in data types that define the kind of values a variable can hold. Below are some of the commonly used data types:

1) Numeric Types:

- int : Integer Values
Example : $x = 10$
- float : floating-point (decimal) Values
Example : $price = 99.99$
- Complex : Complex number
Example : $z = 3 + 4j$

5) Object - Oriented :

Python Supports object-oriented programming (oop) concepts such as classes, inheritance and encapsulation

Differentiate between Interactive Mode and Script Mode with Example.

Here is the difference between Interactive Mode and Script Mode in Python along with Example.

1) Interactive Mode :

- Definition : A mode where Python commands are executed immediately one line at a time
- Used for : Testing small code snippets or individual statements
- Accessed via : Python shell or terminal (`>>>` Prompt).

Example :

```
>>> Print ("Hello")
```

```
Hello
```

```
>>> 5 + 3
```

```
8
```



5 Marks Questions

1) Explain the features of python programming language.

Here are the Key features of the python programming language:

1) Simple and Easy to Learn:

Python has a clear and readable syntax, making it beginner-friendly and easy to understand.

2) Interpreted Language:

Python code is executed line by line, which simplifies debugging and makes development faster.

3) High-level Language:

Python abstracts complex details of the computer, so programmers can focus on logic rather than hardware.

4) Dynamically Typed:

You don't need to declare data type explicitly; python automatically detects the type of variable at runtime.

2) String Type:

- Str: A sequence of characters (texts) enclosed in quotes.

Example: name = "Alice"

3) Boolean Type:

- bool: Represents True or false

Example: is_active = True

4) Sequence Types:

- list: Ordered, mutable Collection

Example: fruits = ["apple", "banana", "cherry"]

- tuple: Ordered, immutable Collection

Example: dimensions = (10, 20, 30)

5) Set Type:

- set: Unordered Collection of Unique items

Example: Unique numbers = {1, 2, 3, 4}

7) What is indentation in python?

Indentation in python refers to the spaces or tabs used at the beginning of a line to define the block of code.

8) Write the syntax of input() and print() functions.

* input() function:

Used to take input from the user as a string

Syntax:

Python

Variable = input("Prompt message")

* Print() Function:

Used to display output on the screen.

Syntax:

Python

print(Object(s), sep=' ', end='\n')

9) Mention any two types of operators in python

1) Arithmetic Operators

2) Comparison (Relational) Operators.

Examples of Python Keywords:

1. `if` - Used for Conditional Statements.
2. `def` - Used to define a function.

4) Define identifiers in Python.

Identifiers in python are the names used to identify Variables, functions, classes, modules, and other objects.

5) List any two limitations of python.

Here are two limitations of python:

1. Slower Execution Speed
2. High Memory Usage

6) Name any two python data types.

Two common python data types are

- 1) `int` - Used to represent integer values (eg: 10, -5)
- 2) `str` - Used to represent String (text) values (eg: "Hello", "Python")



2) Script Mode

- Definition: In this mode, Python code is written and saved in a file with a .py extension, then executed.

- Used for: Writing and running larger programs

- Executed via: A code editor or command line.

Example:

```
name = "Python"
print("Hello", name)
```

3) Discuss the rules for naming identifiers in python with Valid and invalid example.

Rules for naming identifiers in python:

1) Can contain:

- Letters (A-Z, a-z)
- Digits (0-9)
- ** Underscore

2) Cannot start with a digit:

Valid: name1

Invalid: 1name